UNIT 2 · TOPICS 2.3–2.7 · PROBABILITY RULES

Probability: Simulation, Rules & Independence

Estimating probability by simulation, the complement and union rules, mutually exclusive events, conditional probability, the general multiplication rule, and the three equivalent tests for independence — with the distinctions the free-response questions reward.

1. Simulation, Outcomes and the Law of Large Numbers

A **random process** produces results determined by chance; one repetition is a **trial**, its result is an **outcome**, and a collection of outcomes is an **event**. A simulation models the process so that simulated outcomes closely match the real ones.

FOUR-STEP SIMULATION RECIPE

$$\hat{P}(\text{event}) = (\text{number of trials with the event}) \,/\, (\text{total number of trials})$$

1. Assign digits / random numbers to outcomes so their probabilities match the real process. 2. Define one trial (including the stopping rule). 3. Repeat many independent trials and count. 4. Use the relative frequency as the estimate, and state that more trials give a more stable estimate. The **law of large numbers** says this long-run relative frequency approaches a single value (the true probability) as the number of independent trials grows.

CALCULATOR COMMANDS FOR SIMULATION

`rand()` for one uniform value in [0,1), `rand(500)` for a list of 500, `randInt(1,6,500)` for 500 die rolls, `randBin(n,p,500)` for 500 draws from a Binomial count, `randSamp(List,#Trials,1)` for sampling without replacement. Set `RandSeed` first if the run must be reproducible.

2. Sample Spaces, Complements and Basic Probability

FOUNDATIONAL RULES

$$0 \le P(E) \le 1 \quad P(S) = 1 \quad P(E^c) = 1 - P(E)$$

$$P(E) = (\#\text{ outcomes in E}) \,/\, (\#\text{ outcomes in S}) \quad \text{when all outcomes are equally likely}$$

The **sample space** S is the set of all possible nonoverlapping outcomes. E^c ("not E") is the complement. "At least one" is almost always fastest as $1 - P(\text{none})$.

3. Unions, Intersections and Mutually Exclusive Events

ADDITION (UNION) RULE

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$P(A \cap B)$ is the **joint probability** (both happen). "A or B" on the exam is *inclusive*: it includes the overlap. If A and B are **mutually exclusive** (disjoint), they cannot occur together, so $P(A \cap B) = 0$ and the rule collapses to $P(A \cup B) = P(A) + P(B)$.

SYMBOL	READ AS	MEANING
$A \cup B$	A or B	A, B, or both occur (inclusive "or")
$A \cap B$	A and B	Both occur; the joint probability
A^c	not A	A does not occur; complement
$A \mid B$	A given B	A occurs, restricted to the world where B occurred

4. Conditional Probability and the General Multiplication Rule

CONDITIONING AND MULTIPLYING

$$P(A \mid B) = P(A \cap B) / P(B) \quad [P(B) > 0]$$
$$P(A \cap B) = P(A) \cdot P(B \mid A) = P(B) \cdot P(A \mid B)$$

Conditioning **shrinks the denominator** from the whole sample space to B. On a **tree diagram**, the first-level branches are unconditional probabilities and every second-level branch is a conditional probability; multiply along a path to get a joint probability, then add the paths that make up the event.

5. Independence, Tree Diagrams and Exam Traps

THREE EQUIVALENT TESTS FOR INDEPENDENCE

$$P(A \mid B) = P(A) \quad P(B \mid A) = P(B) \quad P(A \cap B) = P(A) \cdot P(B)$$

Independence means knowing whether one event occurred does **not** change the probability of the other. Any one of the three equalities establishes it; a single counter-example disproves it. The special multiplication rule $P(A \cap B) = P(A)P(B)$ is valid *only* when A and B are independent.

TRAPS THE EXAM ACTUALLY TESTS

- **Mutually exclusive ≠ independent.** Disjoint events with nonzero probabilities are strongly *dependent*: knowing A occurred makes $P(B \mid A) = 0$.
- $\mathbf{P(A \mid B) \neq P(B \mid A)}$ in general — the denominator changes. Confusing them is the classic error with medical testing and base rates.
- **Do not assume independence** from "different" or "unrelated" wording; verify with one of the three equalities, or justify it from random sampling / a 10% condition.
- **Sampling without replacement** makes trials dependent. When the sample is at most 10% of the population, treating them as independent is acceptable on the AP exam.
- **"At least one"**: use $1 - P(\text{none})$, not a sum of overlapping events.
- **Probability is long-run**: a simulation with 20 trials is noisy; more trials give a better estimate (law of large numbers), not a guarantee for the next single trial.